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Physiology: Cardiac Impulse Initiation, Propagation, and Autonomic Regulation

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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- Describe the formation of normal heartbeat
- Explain how the cardiac impulse is conducted throughout the heart and its physiological importance
- Interpret conduction velocity in different parts of the heart and its determinants
- Analyze the sympathetic effect on heart rate, electrical conduction and contractility
- Compare and contrast the parasympathetic effect with sympathetic effect on heart rate, electrical conduction and contractility

Source: Course Syllabus

Cardiac impulse generation (Pacemaking)

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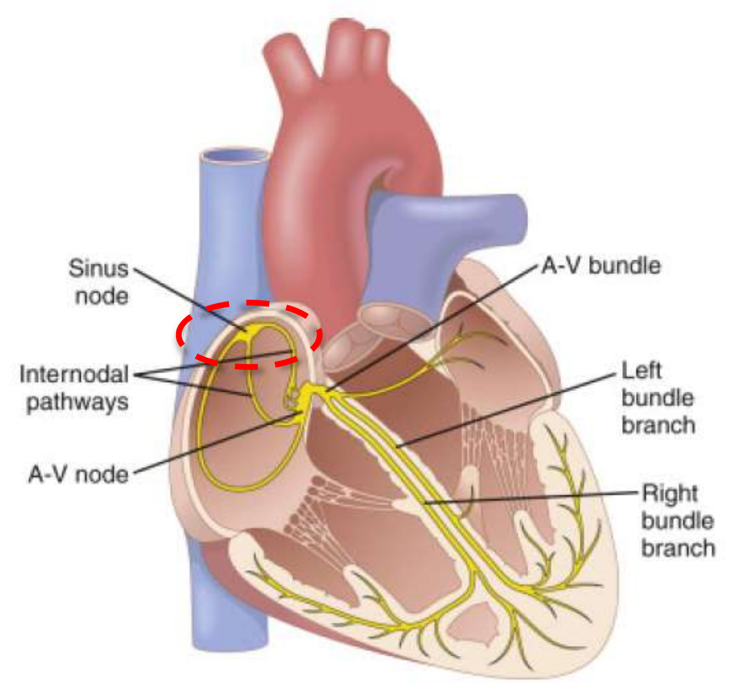

Normal heartbeat is initiated inside the sinoatrial (or sinus) node in the form of slow response action potentials

Guyton and Hall Textbook of Medical Physiology, e13, Figure 10-1

Mechanisms of pacemaking activity

Phase 4 spontaneous depolarization

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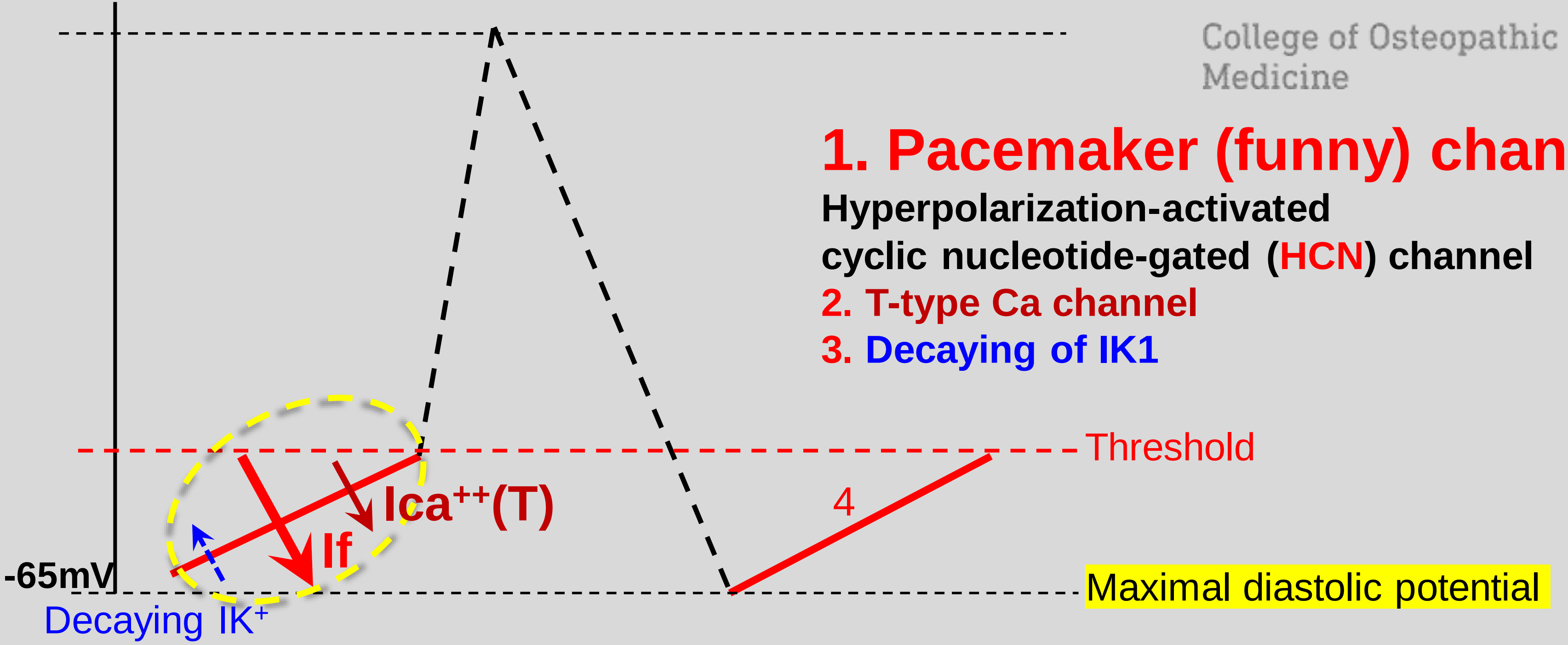

1. Pacemaker (funny) channel

Hyperpolarization-activated cyclic nucleotide-gated (HCN) channel

2. T-type Ca channel

3. Decaying of IK1

- Pacemaker current: Hyperpolarization-activated non-selective cation current (Na^+, K^+) or funny current (If).
- It is different from the fast Na^+ channel seen in fast response fibers

Pacemaking: M-clock & Ca^{++} clock

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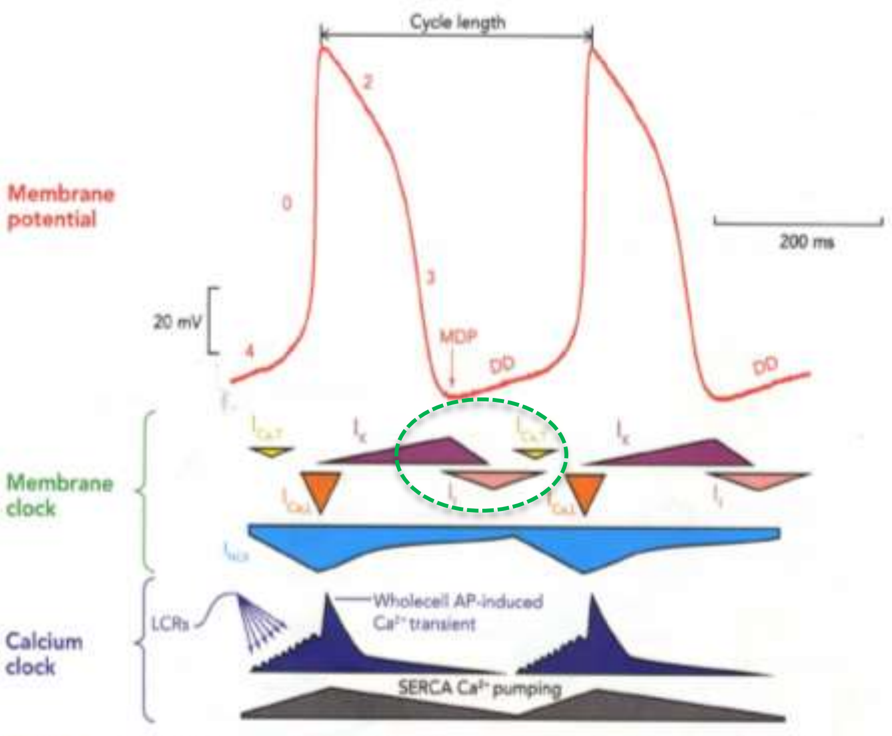
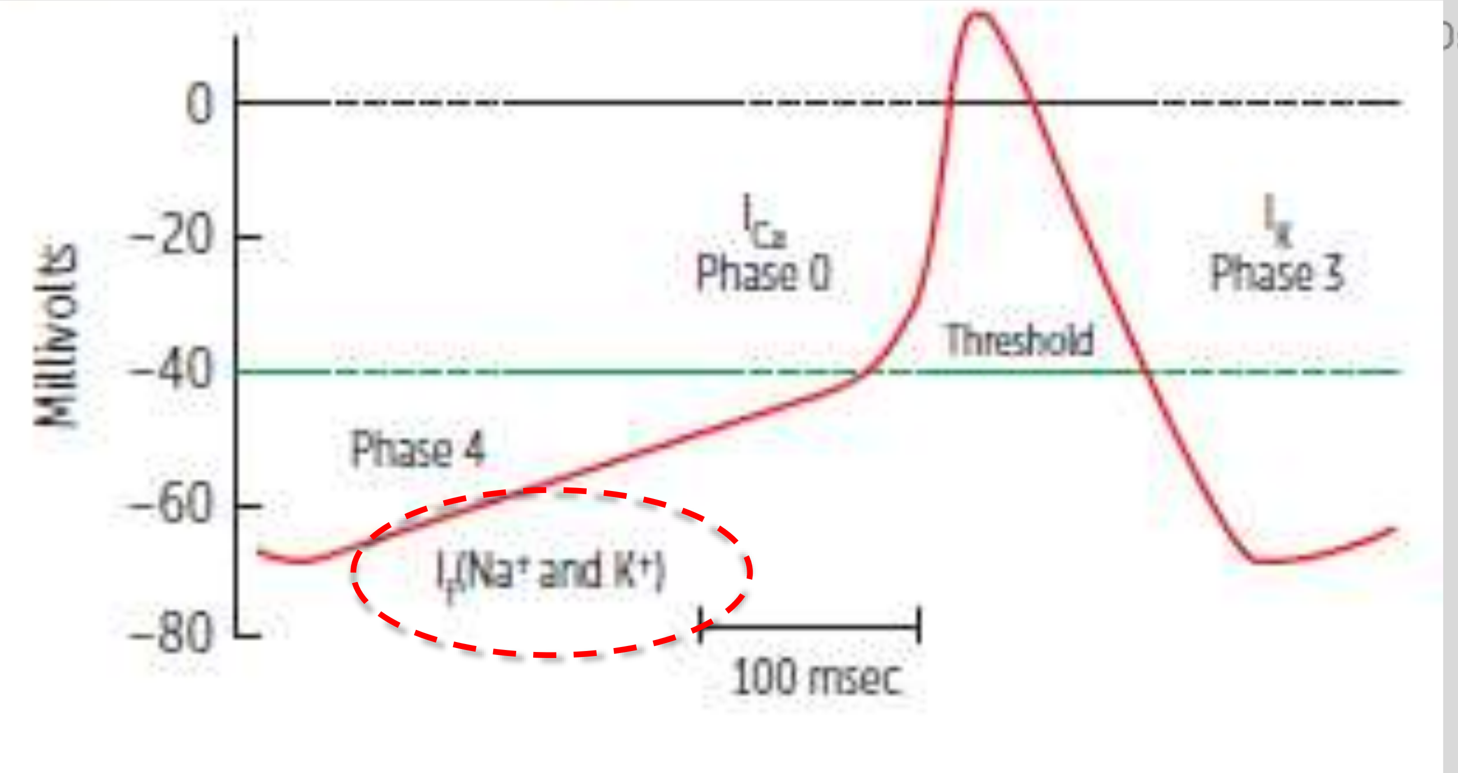


FIGURE 1. A schematic interplay of different membrane clock and calcium clock mechanisms

Besides the 3 channels (I_f, I_{caT} and I_k) collectively termed as membrane (M)-clock (green circle), periodic Ca^{++} release from sarcoplasmic reticulum (SR) also contributes to the pacemaker activity (termed calcium clock)

Physiology 2013;28:74-92

Minimum requirement



USMLE First Aid-Cardiovascular-Physiology

Modulation of pacemaker activity (1)

Speed of depolarization

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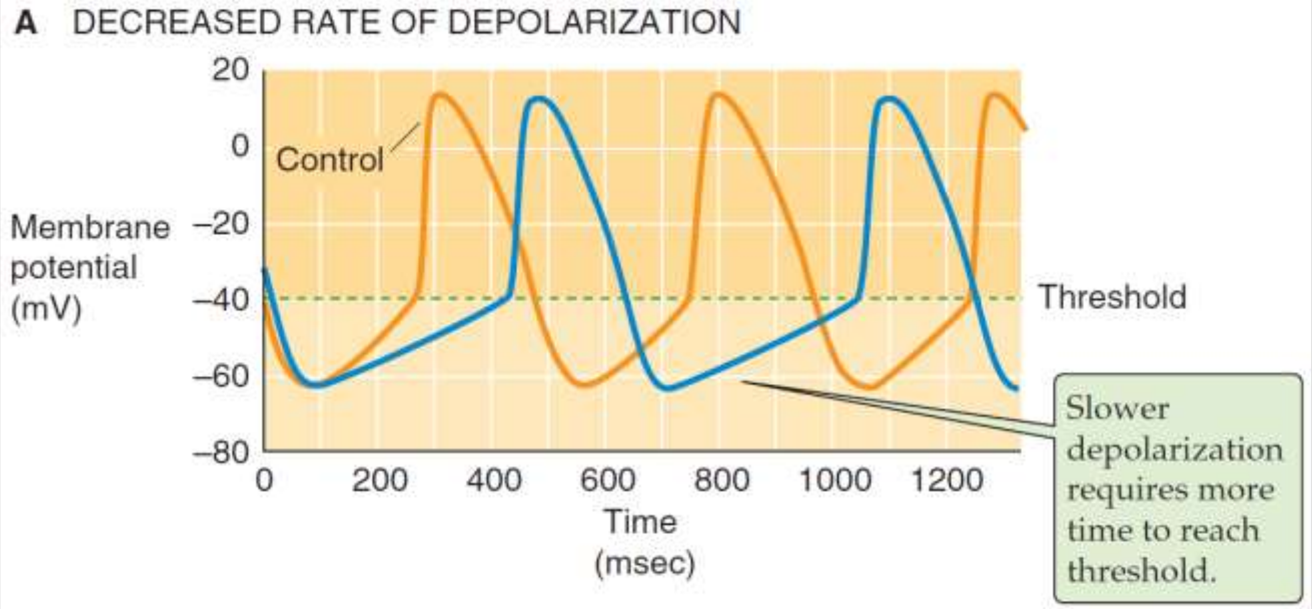

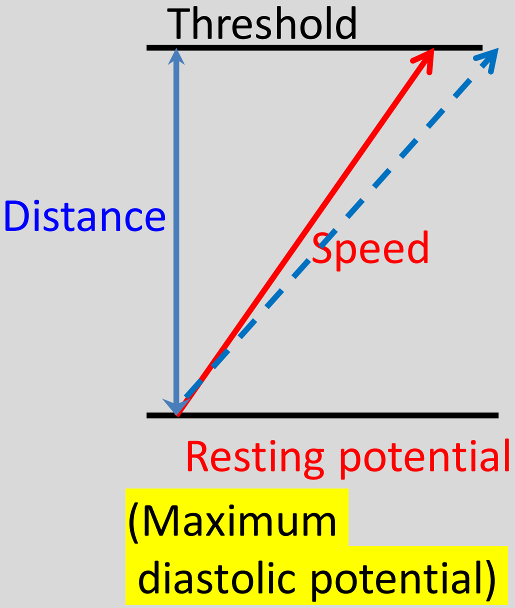

Boron & Boulpaep, Medical Physiology UP Ed, PP492

Modulation of pacemaker activity (2)

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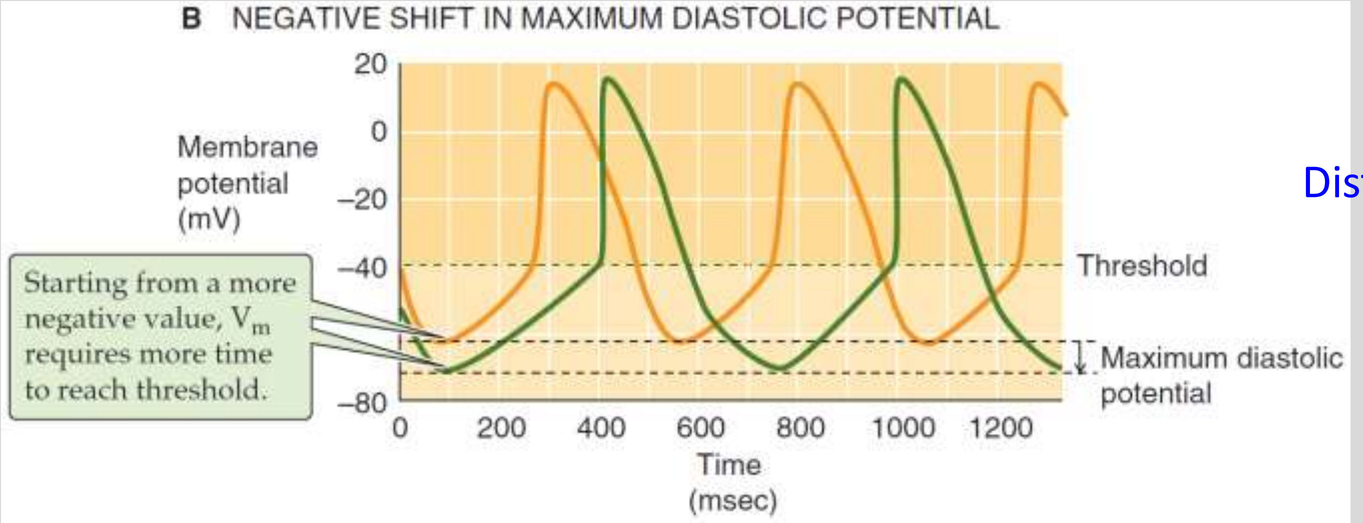
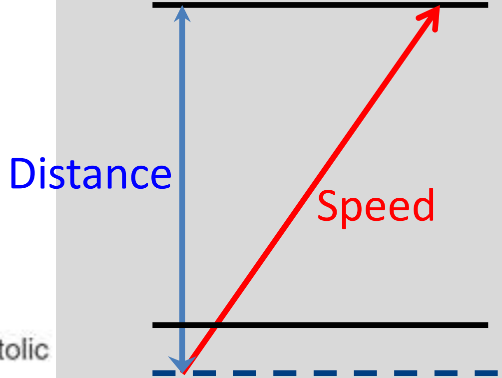
Maximum diastolic potential level





Resting potential (Maximum diastolic potential)

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Modulation of pacemaker activity (3)

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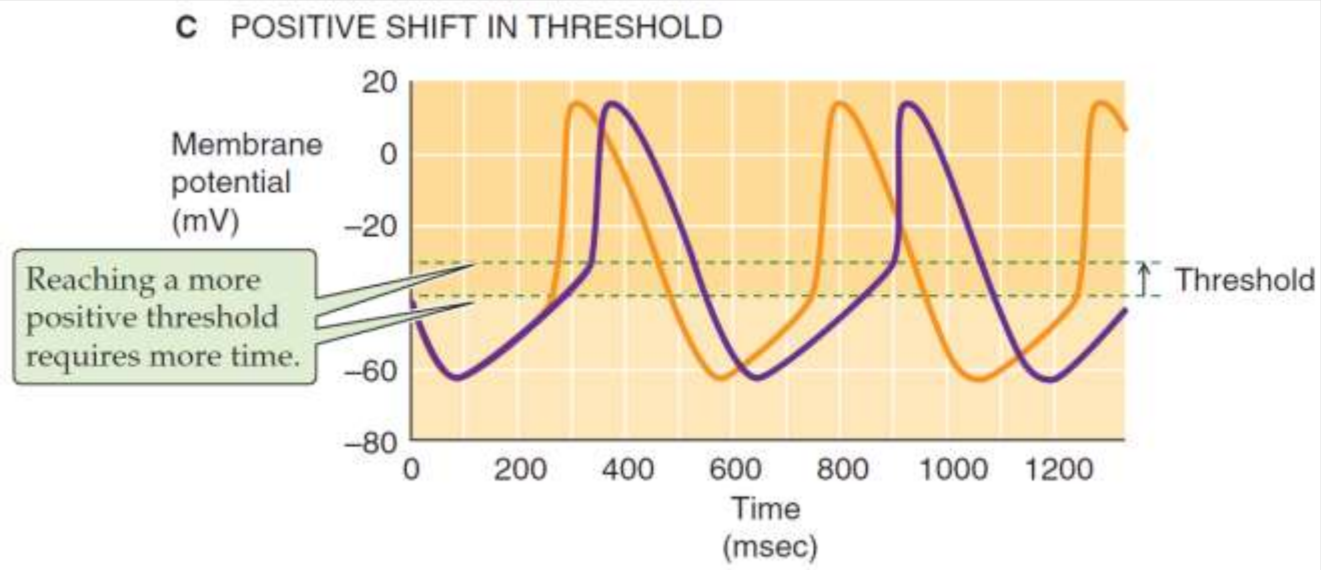
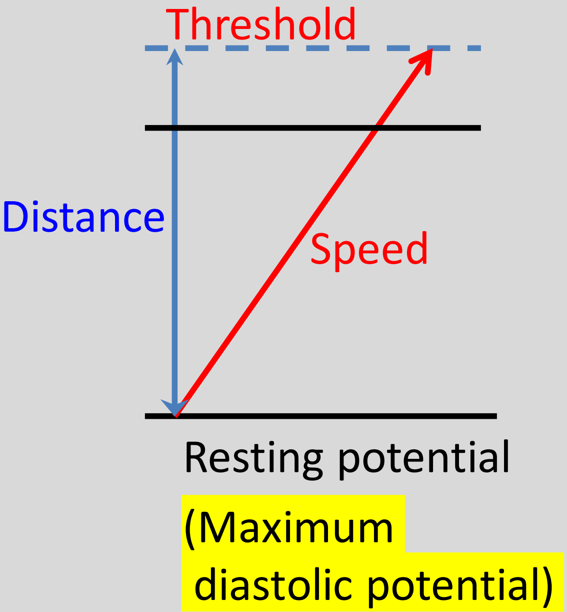
Threshold level





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Cardiac pacemaker hierarchy

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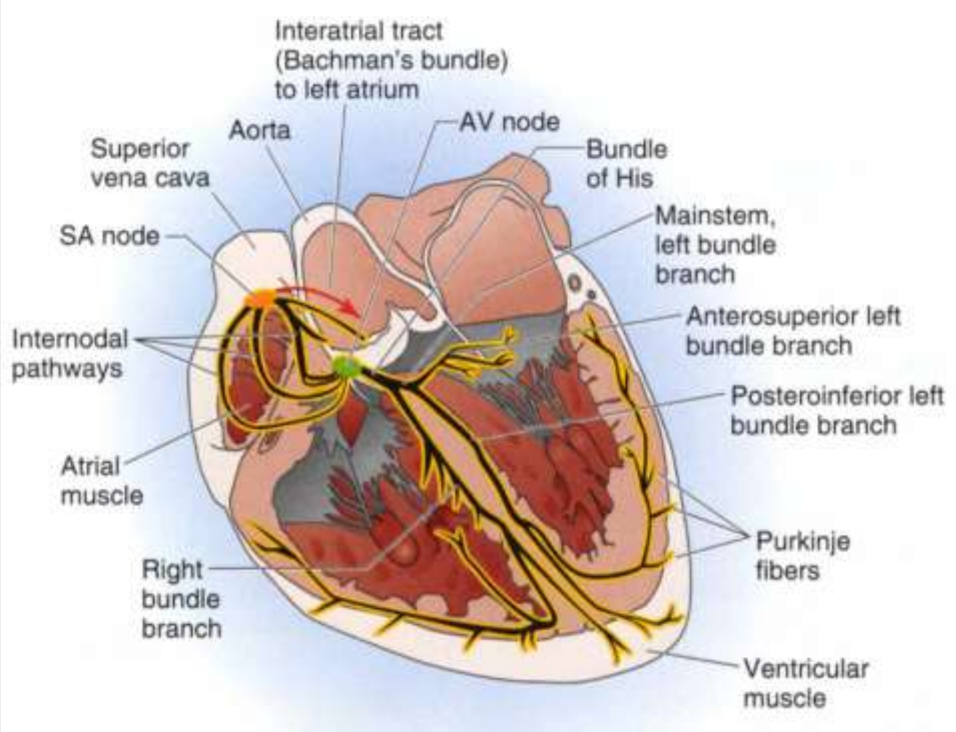

1. **Sinus node:** 60-100bpm (primary)
2. **Atrioventricular node:** 40-60bpm (secondary)
3. **His-Purkinje system:** 20-40bpm (ventricular escape beats)

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Cardiac Impulse Propagation

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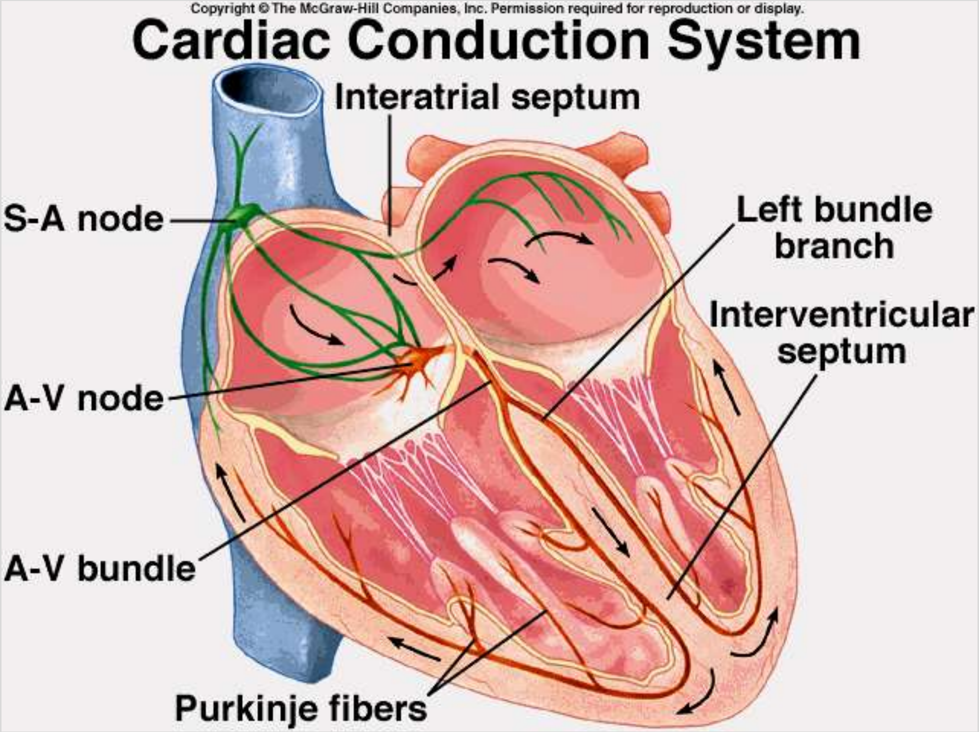

Heart beat (AP) initiated in

SA node → Atria (R-L) →

AV node → His bundle →

Bundle branches (R-L) →

Purkinje fibers → Ventricles

The propagation sequence determines: 1. ECG waveforms

2. Also affects cardiac function

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Sinoatrial node (primary pacemaker)

(SA node or sinus node)

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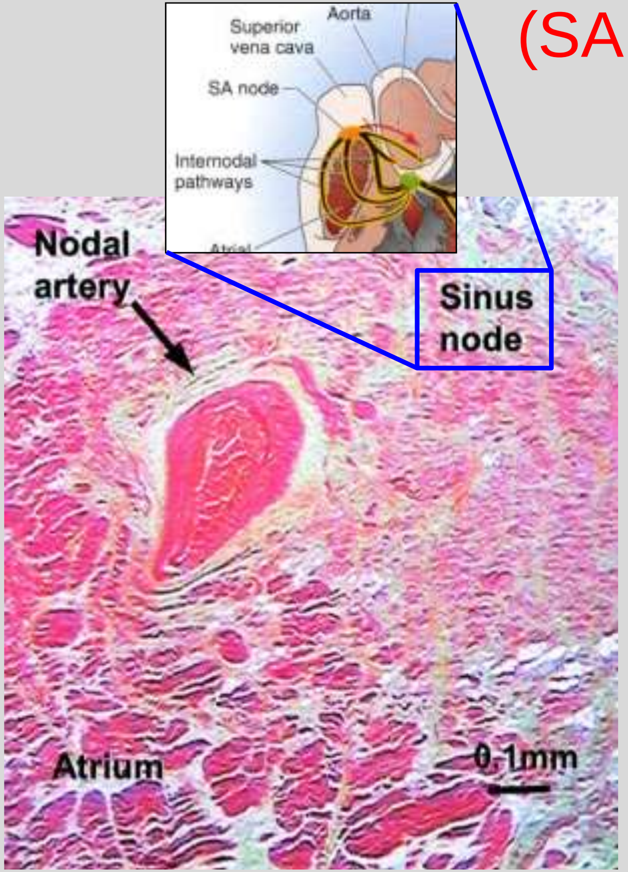

1. Location: Superior vena cava (SVC) & right atrium (RA) junction, or High-RA clinically
    - Center: nodal or p-cells
    - Periphery: transitional cells between nodal and atrial fibers
2. Gap Junction: connects adjacent cardiac cells, allowing ions to pass between cells (sinus node: cx45)
3. Heart rate: 60-100bpm (average 72bpm)
    - >100bpm=tachycardia
    - <60bpm=bradycardia

Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1

Todor N. Mazgalev et al. Circulation. 2001;103:2660-2667

Electrical propagation within atria

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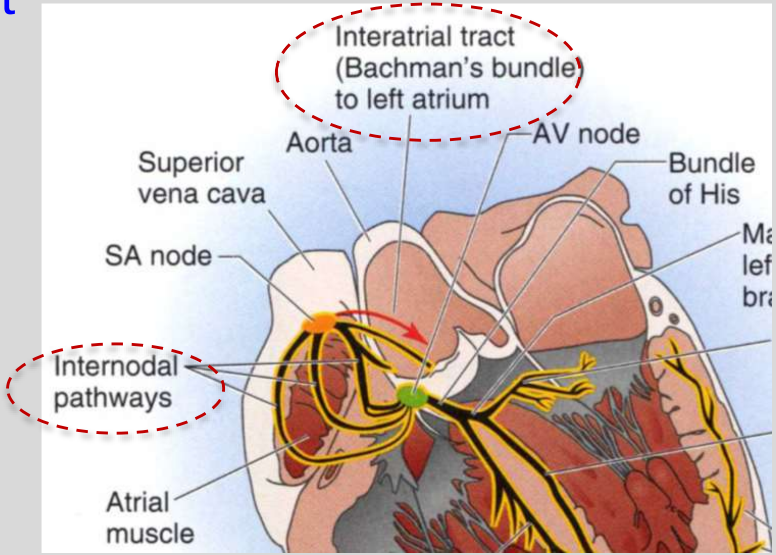
1. From SA node (located at high RA)
2. Going down (top to bottom in RA) through 3 internodal pathways
3. Also from RA to LA via Bachman's bundle (located at the roof of the atria)

- ➢ General direction: Top→bottom, right→left

(atrial depolarization creates P wave on ECG)

- ➢ P wave axis: downward and leftward

Note: Internodal pathways and interatrial tract are muscle bundles (preferential pathways), not specialized cardiac myocytes.



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Longitudinal vs Transverse Conduction

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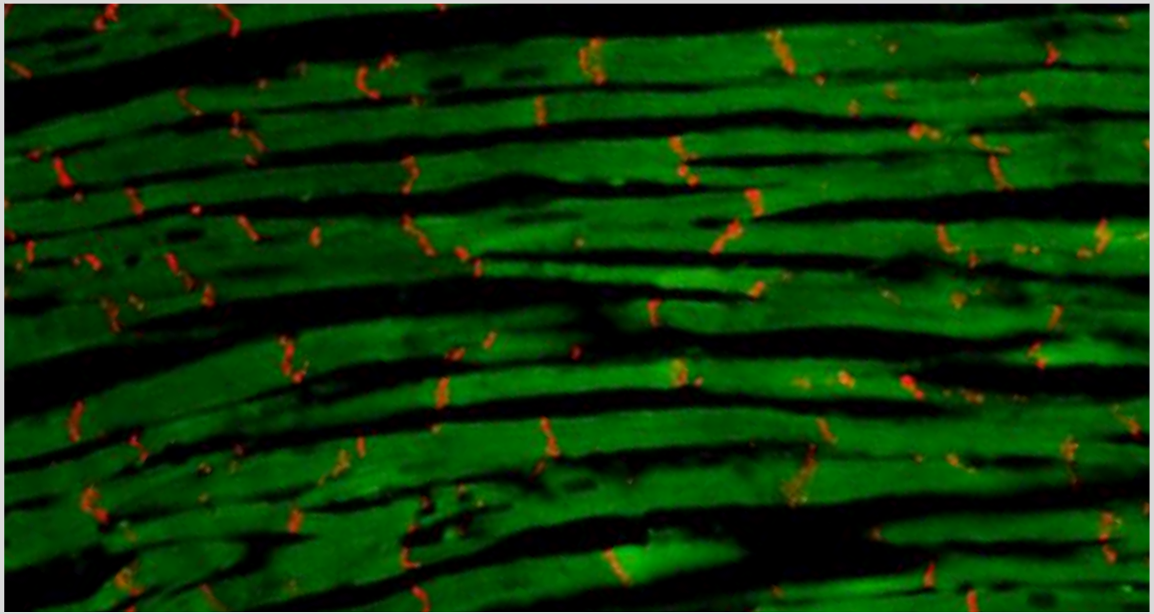
Faster longitudinal conduction (less cells to travel, more gap junctions)



Fast conduction along fiber orientation

Slower transverse conduction (more cells to travel, less gap junctions)

Cx43 (red) in ventricular tissue (our own data)

Gap Junctions

Made of different connexins

(Cxs 40, 43, 45 etc)

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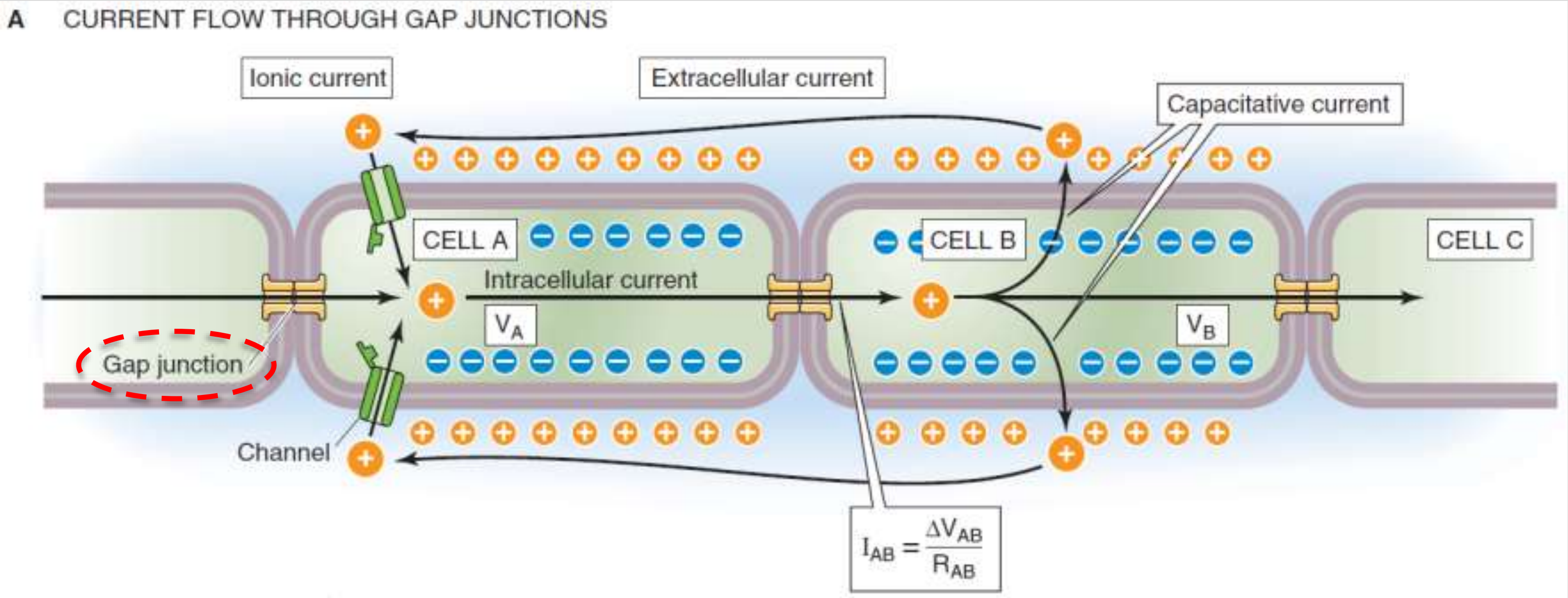
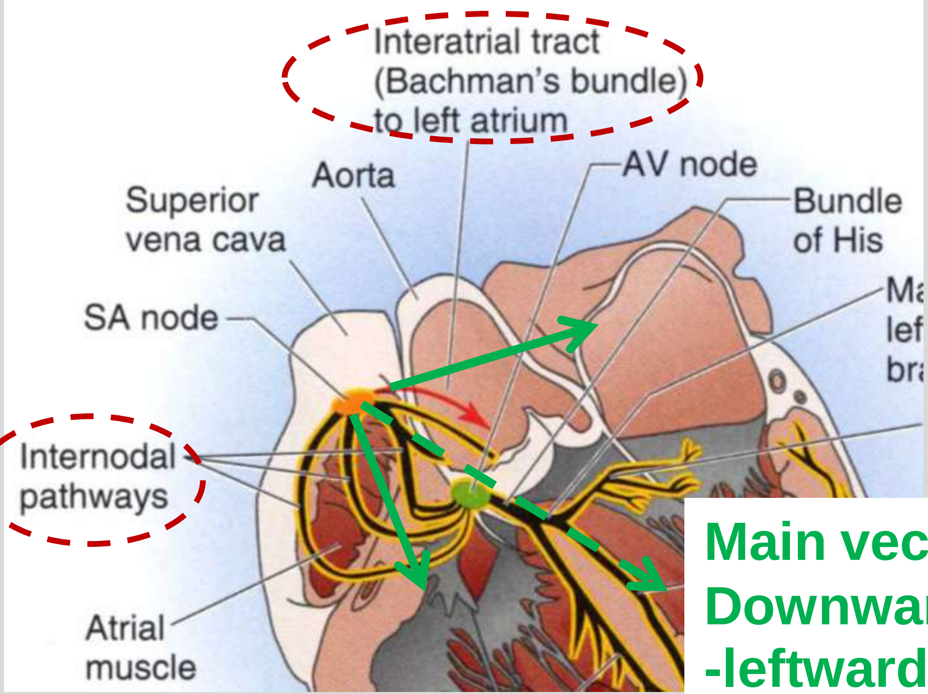
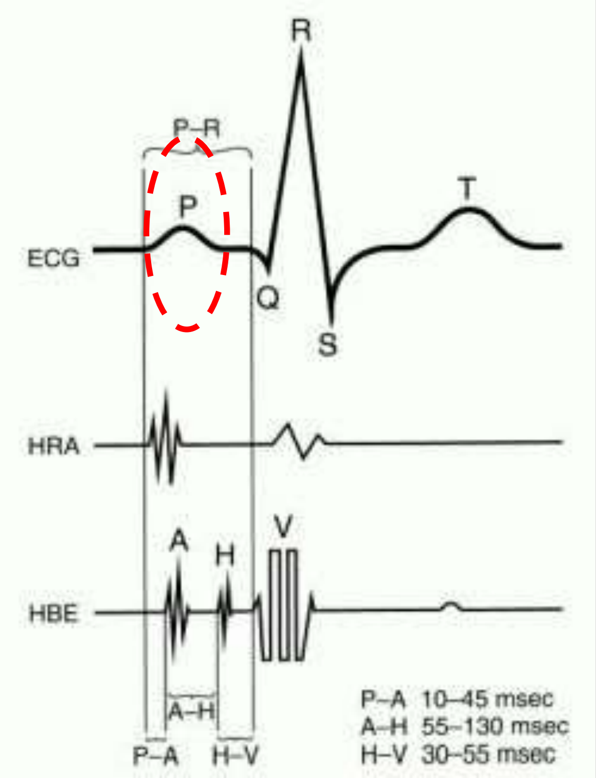


Boron Medical Physiology Figure 21-3A.

Electrical propagation within atria (atrial depolarization creates P wave)



Main vector Downward -leftward



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Atrial excitation and contraction

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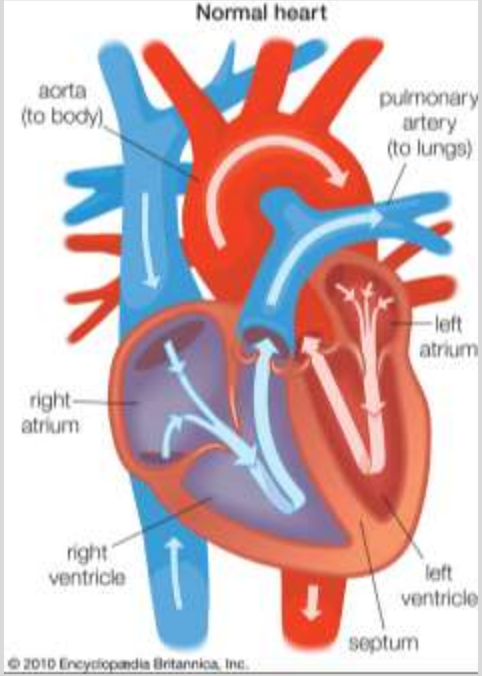
- Excitation of both atria is from top→bottom direction
- Helps to move blood down into the ventricles (lower chambers)



Slow electrical propagation in the atrioventricular node (AVN)

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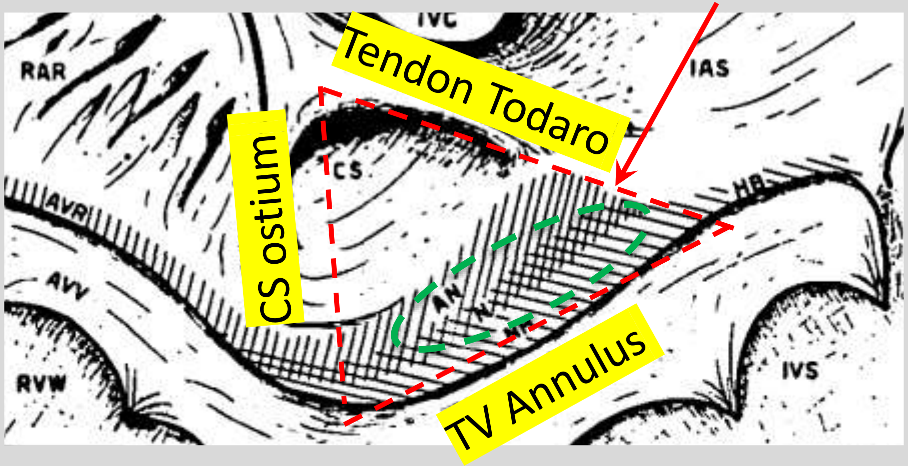
1. Location: lower inter-atrial septum (clinically known as the triangle of Koch)
2. Histology: AN (atrial-nodal), N (Nodal), NH (nodal-His) cells
3. Electrophysiology: a. Creates physiological AV delay due to slow conduction → optimize ventricular filling

b. Secondary (backup) pacemaker

c. Filtering role during very high atrial rates, like atrial flutter or atrial fibrillation (AF).

Triangle of Koch



Not shown on ECG

CS: coronary sinus; TV: tricuspid valve

Paes De Carvalho et al, Circ Res, 1960;8:801-809

Triangle of Koch (viewed from RA)

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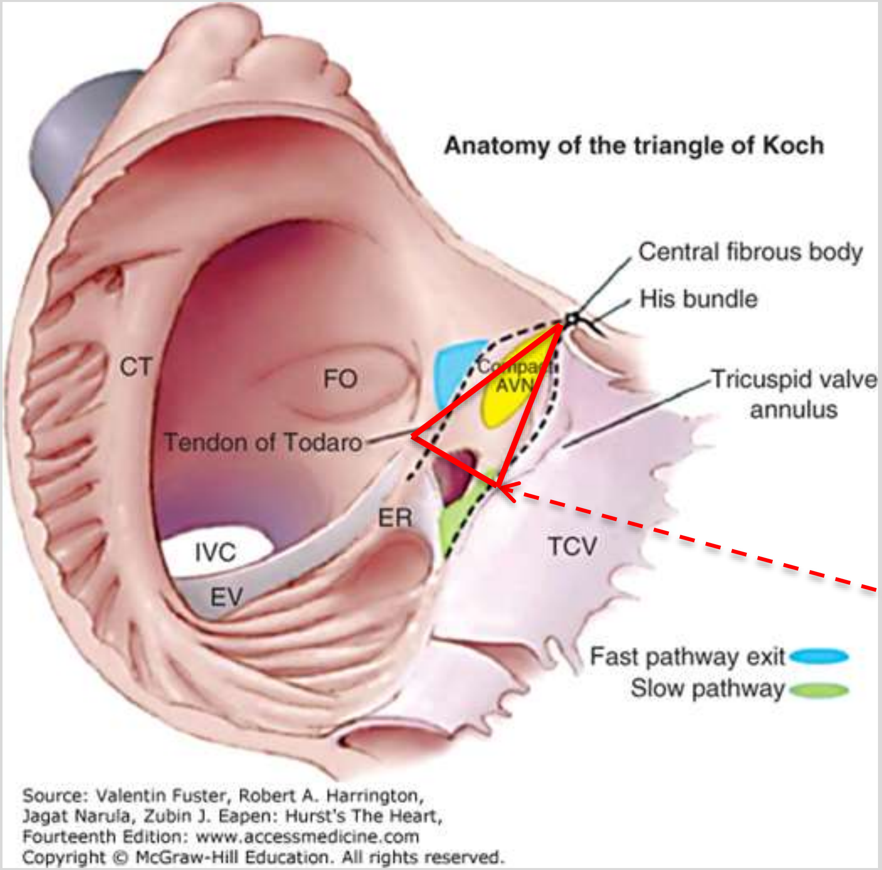

Ostium of coronary sinus

Fig. 78-9 in Hurst's The Heart 14e

Mechanisms of slow conduction in the AV node

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- Slow response AP-I_{Ca++}(L) versus fast I_{Na+}
  Lower dv/dt (maximal speed of depolarization) & lower amplitude
- low-conductance connexin (Cx45)
- Smaller cell size
- Dense connective tissue in the AV node

A sample question: To inhibit AV conduction, which medication would you choose?

L-type calcium channel blocker
or sodium channel blocker?

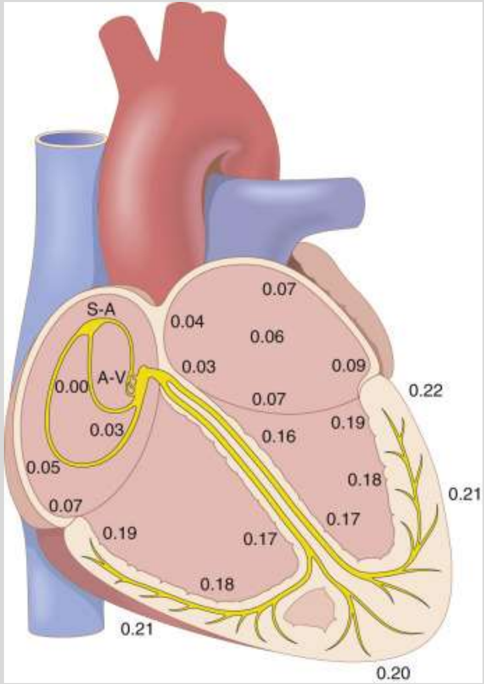
His-Purkinje system

Rapid electrical transmission

- Bundle of His
- Right bundle branch
- Left bundle branch bifurcates into anterior and posterior fascicles

(Insulated in fibrous tissue, no contact with ventricular myocytes)

- Purkinje fibers: distributed sub-endocardially from apex toward base.
  directly activate ventricular myocytes
- endocardial myocytes are activated first (vs epi)
- highest conduction velocity (CV)
- responsible for synchronous ventricular activation



Guyton and Hall Textbook of Medical Physiology Figure 10-4. © 2016

His-Purkinje system & ECG

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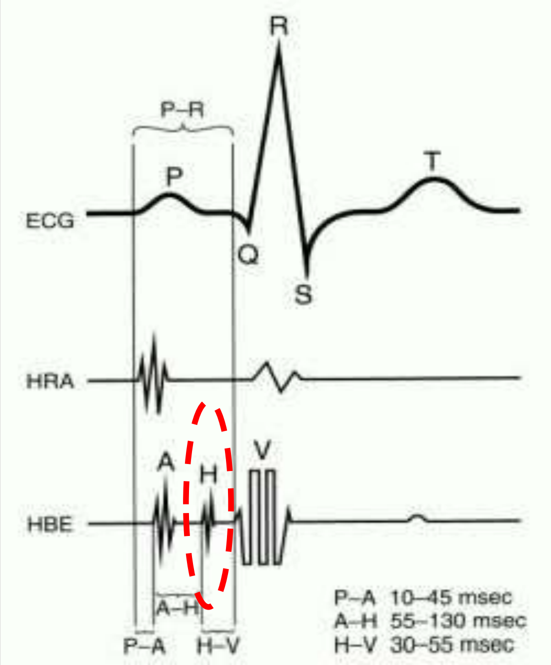
Bundle of His: not shown on ECG,
can be recorded with intracardiac catheter,
as shown in the figure (H, His electrogram)

Bundle branches: not shown on ECG
Bundle branch block can be
diagnosed with ECG, because it alters
normal ventricular activation sequence



Purkinje fibers have the fastest conduction velocity

Mechanisms

1. Fast response AP (fast INa^+)
   high dv/dt (maximal speed of depolarization) & amplitude
2. Largest cell size
3. High conductance connexin (Cx 43)

Ventricular Activation Sequence

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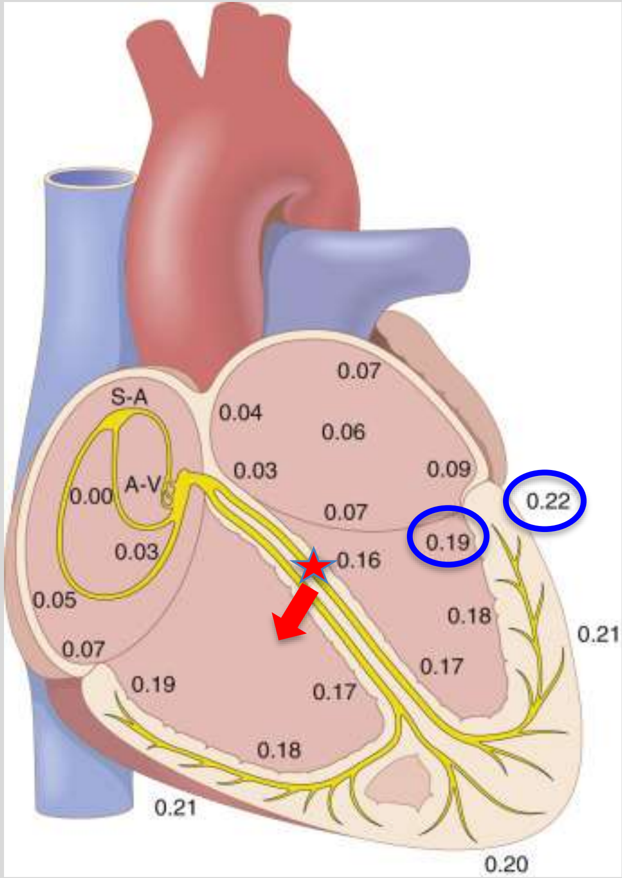

Guyton and Hall Textbook of Medical Physiology Figure 10-4. © 2016

Note: the numbers shown are timing from SA node, in seconds)

Due to a septal fascicle from left bundle, interventricular septum on the left side (★) is the earliest to be excited in the ventricles.

Create a rightward vector

Responsible for septal “q wave” on ECG

Generally from apex to base, endocardium to epicardium

A sample question

- Which side of the ventricles is activated earlier: endocardial vs epicardial? and why?

Depolarization of the ventricles creates QRS complex on ECG

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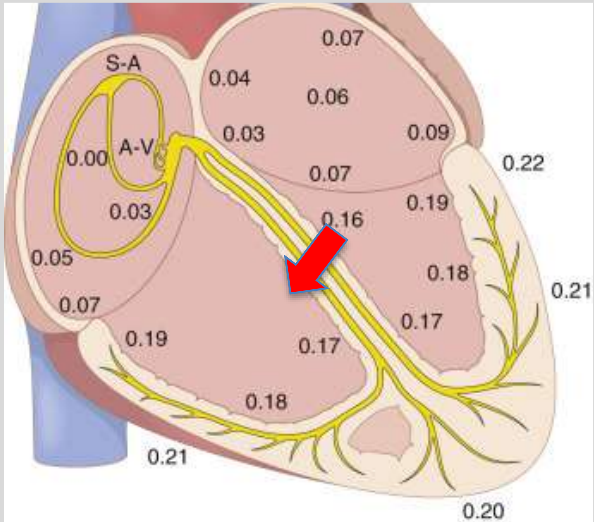
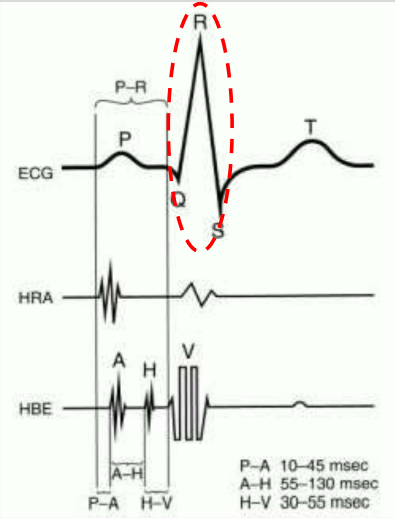
Except the earliest ventricular site at the upper inter-ventricular septum on the left side (which creates a rightward vector, arrow), the remaining general depolarization sequence:

- Apex → base
- Endocardium → epicardium (due to the Purkinje fibers)

Ventricular depolarization creates QRS complex (circled) on ECG.





Guyton and Hall Textbook of Medical Physiology Figure 10-4. © 2016

A sample question

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Which medication can slow ventricular conduction?

L-type calcium channel blocker or

sodium channel blocker?

Normal excitation is important for ventricular Function

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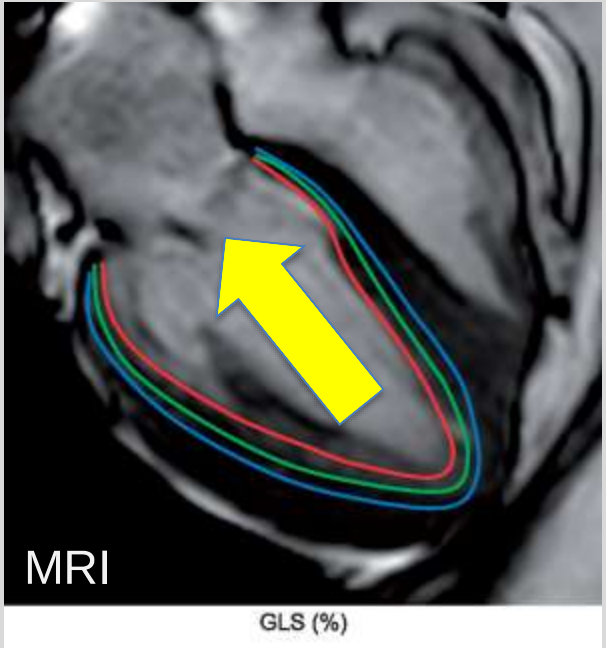
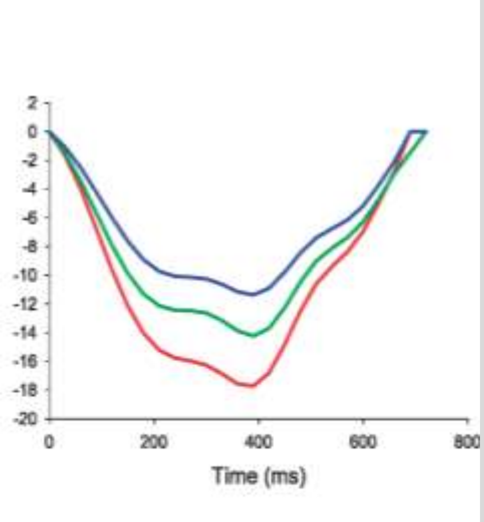
Contraction starts endocardially (inside wall) & from apex toward base (aorta)-pump blood up into aorta





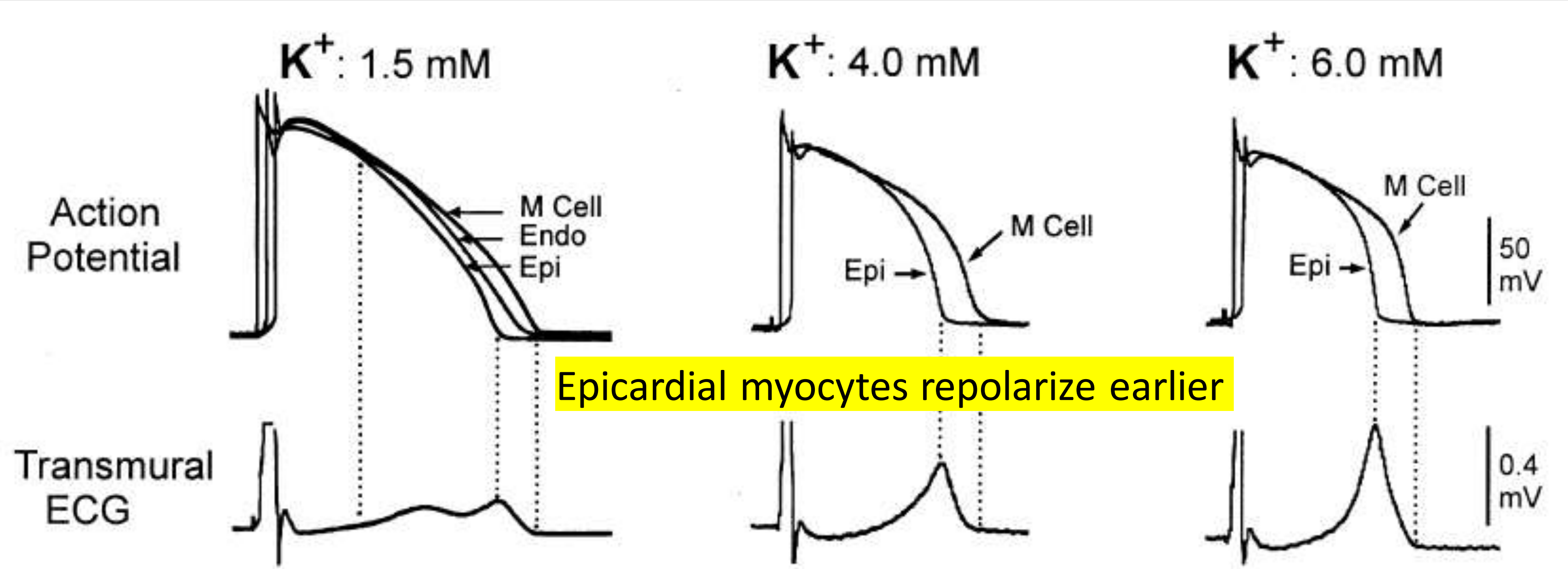
Contraction starts inside and spread to outside

Curr Cardiol Rev. 2017 May; 13(2): 118–129

Eur Heart J Cardiovasc Imaging 2019;20(6): 605–619

Ventricular Repolarization

Starts epicardially (outside wall)
due to shorter action potential duration
in epicardial versus middle/endocardial layer



Epicardial myocytes repolarize earlier

Yan et al. J Am Coll Cardiol 2003;42:401–9

Ventricular repolarization sequence

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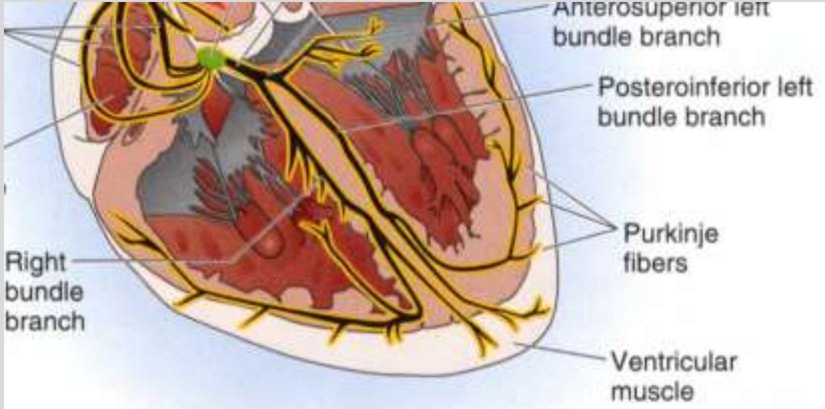
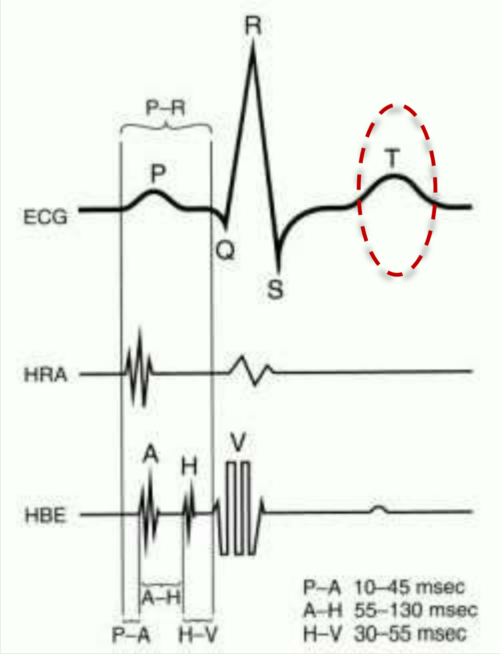
1. Epicardium → endocardium
   (opposite to depolarization)
2. Apex → base
   (same as depolarization)
3. Creates T wave on ECG
   (normally follows QRS polarity)

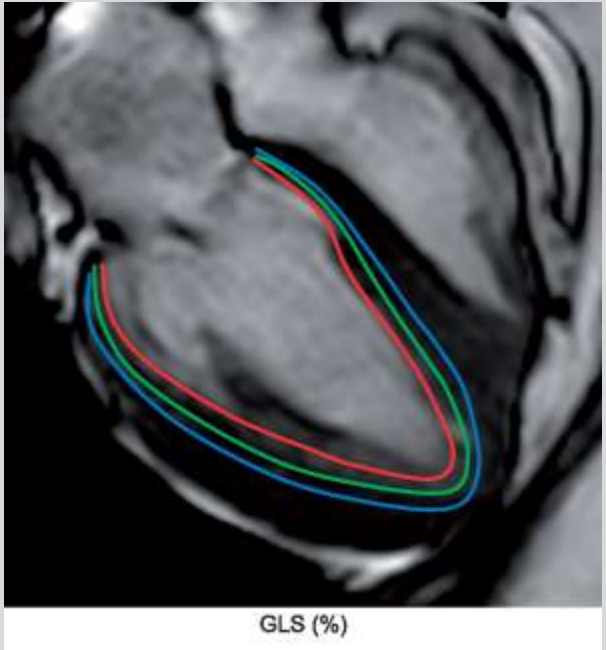




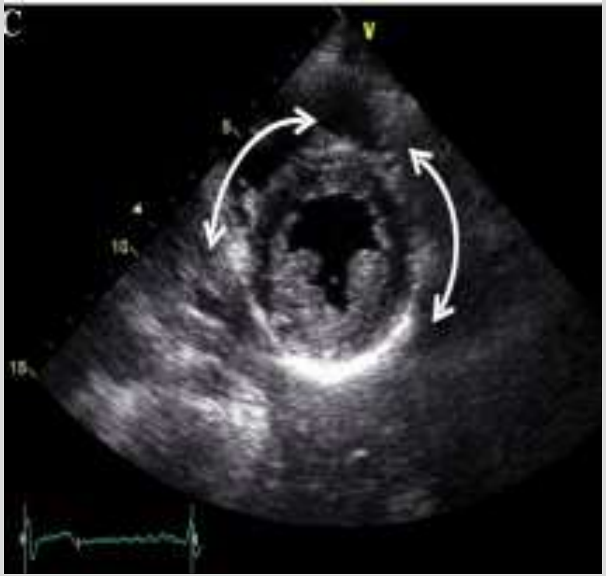
Boron & Boulpaep Medical Physiology UP Ed PP484, Figure 20-1

Functional importance

Relaxation starts at apex, epicardially (outside wall) to optimize ventricular relaxation



Eur Heart J Cardiovasc Imaging 2019;20(6): 605–619



Curr Cardiol Rev. 2017 May; 13(2): 118–129

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Conduction Velocity (CV)

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- CV: the speed at which APs are propagated (unit: m/sec).
- CV differs in different parts of the heart (table):
- The propagation of APs depends on the spread of local currents generated by the APs, correlated with the size of the inward current during the upstroke, or the rate of rise of the AP (dV/dt).
- Other factors: cell size, AP magnitude, gap junctions (types of connexins 43, 40, 45), fibrosis, etc.

Sequence	Structure	CV (m/sec)
1	SA node	0.01
2	Atria	0.5-1
3	AV node	0.01-0.05
4	His Bundle	2
5	Bundle branches	2
6	Purkinje fibers	2-4
7	Ventricles	0.5-1

Katz AM. Physiology of the Heart, third edition. Philadelphia: Lippincott, Williams, and Wilkins, 2001.

Minimum requirement

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- Speed of conduction—Purkinje > atria > ventricles > AV node.
- Pacemakers—SA > AV > bundle of His/Purkinje/ventricles.
- Conduction pathway—SA node-atria-AV node-common bundle-bundle branches-fascicles-Purkinje fibers-ventricles.
- SA node “pacemaker” inherent dominance.
- AV node—located in posteroinferior part of interatrial septum.
  Blood supply usually from RCA.
  100-msec AV delay allows time for ventricular filling.

Effect of Autonomic Nervous System on Heart

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	Parasympathetic (Vagal)	Sympathetic
Sinus rate (Chronotropic)	↓	↑
AV Conduction (Dromotropic)	↓	↑
Contractility (Inotropic)	↓	↑

Vagal stimulation on heart rate

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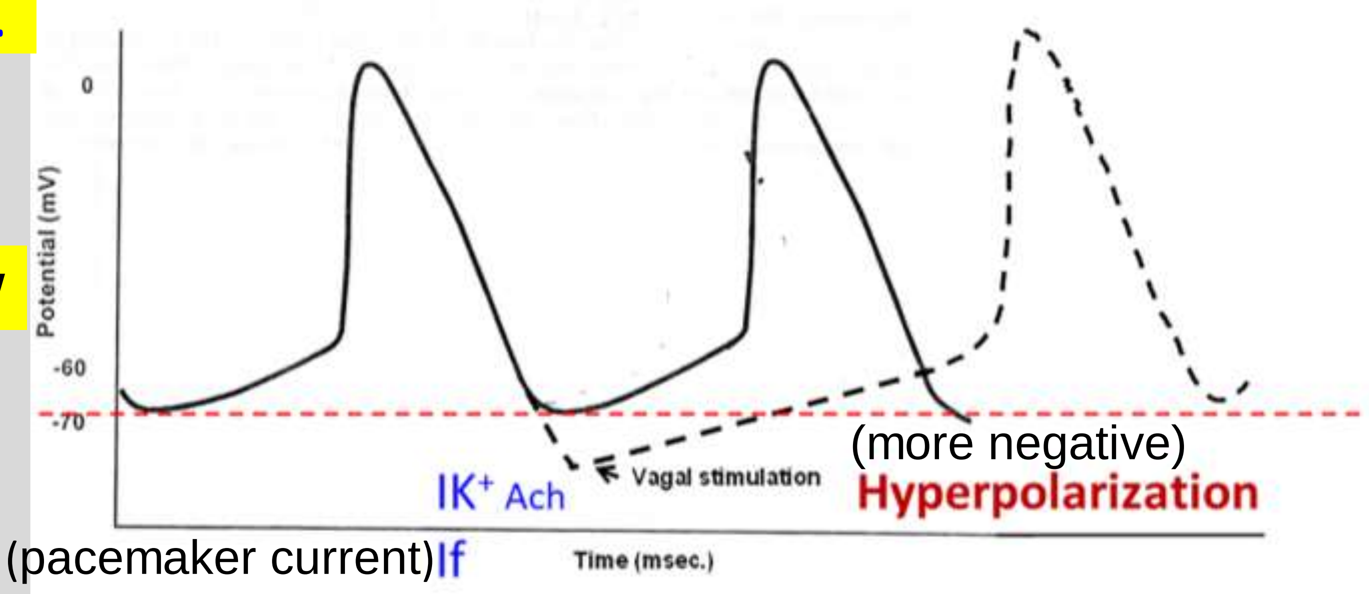
Neurotransmitter: Acetylcholine

Receptor: Muscarinic (M) receptor

1. Ligand gated K+ Channel → hyperpolarization
2. ↓If, $I_{Ca\text{-}L}$, etc.

Inhibit or slow heart rate (HR)



Note: Strong VS inhibits SA and AV nodes, resulting in ventricular escape beat

Signaling pathway mediating vagal effects

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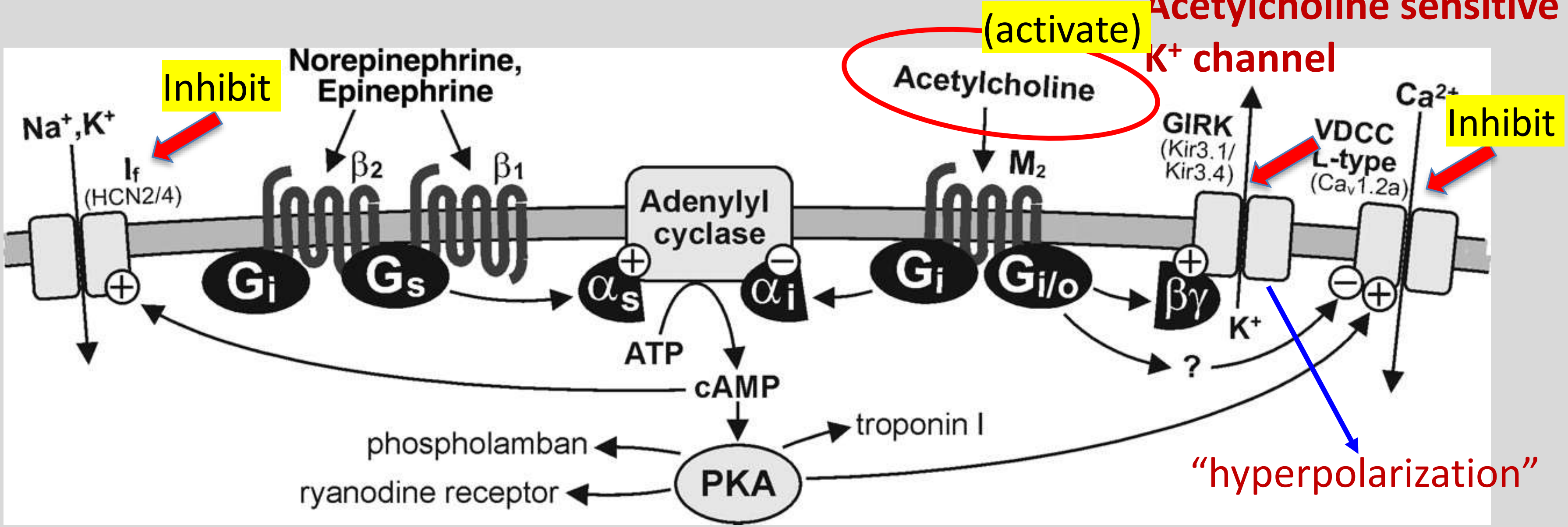
Decrease HR: Negative chronotropic effect, ↓If

Decrease AV conduction: Negative dromotropic effect,↑Ik_{ach} ↓Ica^{++}-L

Decrease myocardial contractility: Negative inotropic effect, ↓Ica^{++}-L



Physiol Rev 2005;85:1159-1204

Sympathetic effects on pacemaker activity

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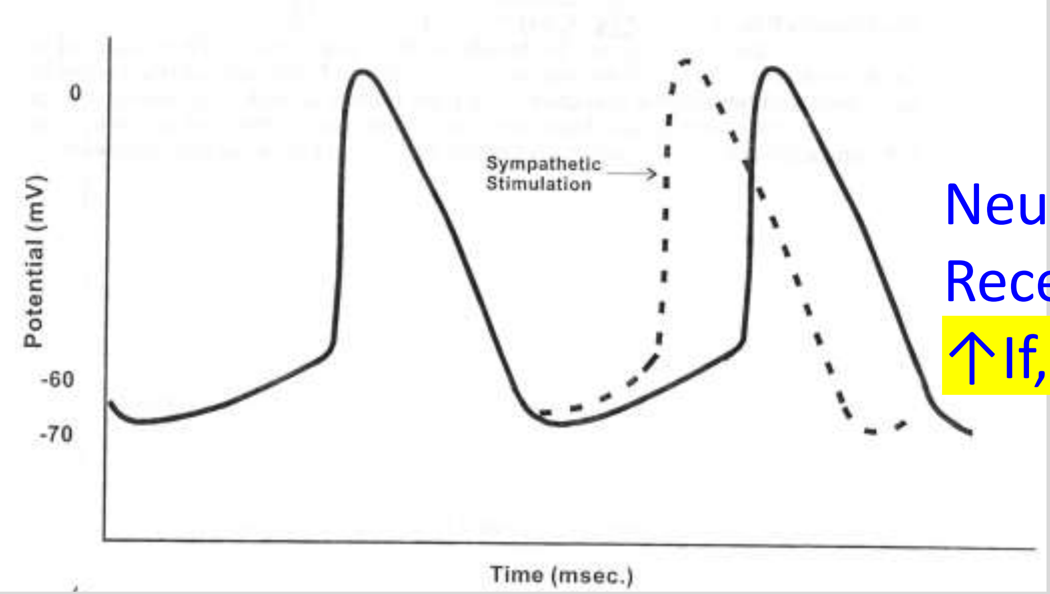
Mediating "The fight-or-flight response "



Neurotransmitter: norepinephrine

Receptor: Adrenergic β1-receptor

↑If, $I_{Ca\text{-}L}$, etc.

Maximum Heart Rate ≈ 220bpm-age (years)

(under maximal sympathetic stimulation)

Signaling pathway mediating sympathetic effects

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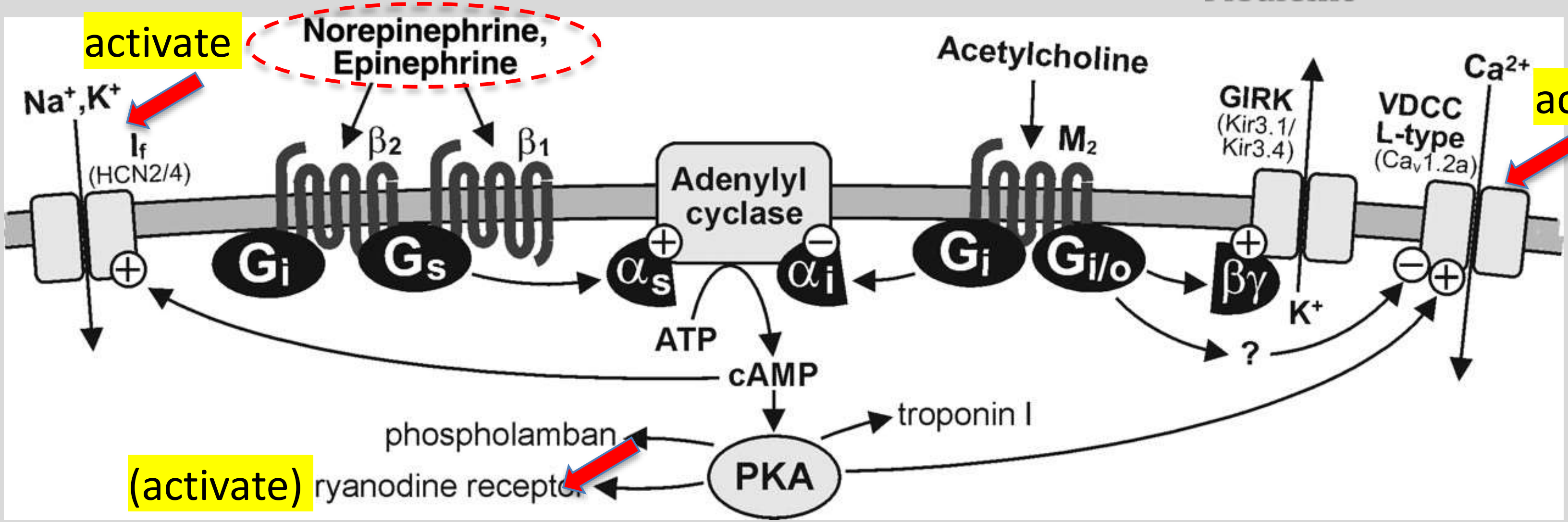


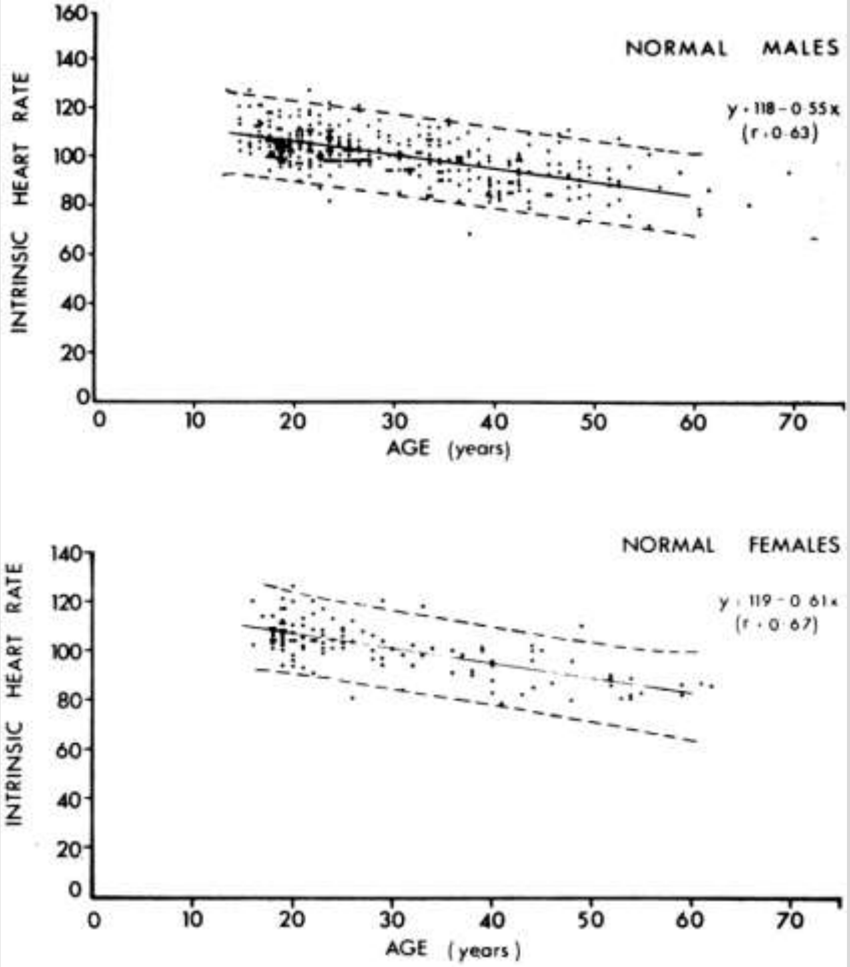
Increase heart rate: Positive chronotropic effect (If)

Increase AV conduction: Positive dromotropic effect (Ica^{++})

Increase myocardial contractility: Positive inotropic effect (Ica^{++}, Ca release)

Physiol Rev 2005;85:1159-1204

Concept of intrinsic heart rate



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Without autonomic influence

Block both vagal and sympathetic effects

Cardiovasc Res 1970;4:160–167

Summary Slide

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- Pacemaker hierarchy
- Atrial propagation
- AV node conduction
- His-Purkinje system
- Ventricular activation
- Vagal effect: HR, AV conduction, contractility
- Sympathetic effect: HR, AV conduction, contractility
- Core References: Guyton and Hall Textbook of Medical Physiology, 14e, Chapter 10-title: Rhythmical Excitation of the Heart.

  Boron, Boulpaep. Medical Physiology: A Cellular And Molecular Approach 2e. 2012, Elsevier.

Lecture and Lab Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R